

## Final Review

Please be sure to look at past slides and exercises for review as well.

### Semantics – Set Theory

#### Important to remember:

1. Definition: A set is an abstract collection of distinct objects.

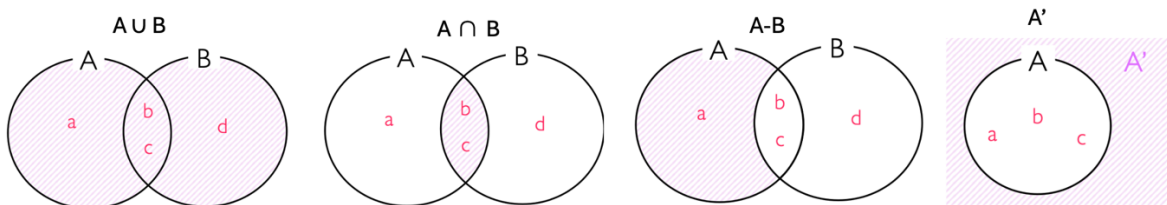
- Important: objects in a set must be distinct. two of the same objects in a set counts as one – e.g.  $\{a, a, b\} = \{a, b\}$ .
- Note that sets can also be objects in a set – a set of sets.

2. Some important notions about sets:

- **Membership:** We write  $a \in S$  if  $a$  is a member of the set  $S$ , and  $a \notin S$  otherwise.
  - Example: suppose  $S = \{a, b, c\}$ . Then  $a \in S$ , and  $d \notin S$ .
- **Identity:** set  $A$  and set  $B$  are identical/equal iff they have exactly the same members, written as  $A = B$ . If  $A$  and  $b$  are not identical/equal, we write  $A \neq B$ .
  - Example: let  $A = \{a, b, c\}$ ,  $B = \{a, b, c\}$ . Then  $A = B$ .
- **Subset:** set  $A$  is a subset of set  $B$  iff every element of  $A$  is also an element of  $B$ , written as  $A \subseteq B$ . If  $A$  is not a subset of  $B$ , we write  $A \not\subseteq B$ .
  - Example: let  $A = \{a, b, c\}$ ,  $B = \{a, b, c, d, e\}$ . Then  $A \subseteq B$ .
- **Cardinality:** denoted by  $||$ , indicates the number of elements in a set.
- Note: there are also two special sets:
  - The empty set: The set with no members. Written as either  $\emptyset$  or  $\{\}$ .
  - The universe of discourse: The set containing everything. Written as  $U$ .

3. Operations we can do on sets:

- **Union:** written as  $A \cup B$ . The union of  $A$  and  $B$  simply adds the elements in each set together.
  - e.g., let  $A = \{a, b, c\}$ ,  $B = \{b, c, d\}$ . Then  $A \cup B = \{a, b, c, d\}$ .
- **Intersection:** written as  $A \cap B$ . The intersection of  $A$  and  $B$  singles out only the elements that are found in both sets.
  - e.g., let  $A = \{a, b, c\}$ ,  $B = \{b, c, d\}$ . Then  $A \cap B = \{b, c\}$ .
- **Difference:** written as  $A - B$ .  $A - B$  picks out elements that are only in  $A$  and not in  $B$ .
  - e.g., let  $A = \{a, b, c\}$ ,  $B = \{b, c, d\}$ . Then  $A - B = \{a\}$ .
- **Complement:** The complement of set  $A$  is written as  $A'$ . The complement of  $A$  is the difference between the set containing everything and  $A$ .
  - e.g., let  $A = \{a, b, c\}$ . Then  $A'$  is the set that contains anything but  $a, b$ , and  $c$ .



4. Ways to define a set:

- **Venn diagrams**
- **List notation:** by listing all of its members
  - Chenli's cat friends =  $\{\text{Remy, Bailey, January}\}$

- **Predicate/Abstract notation:** by stating a condition that is satisfied by all and only the elements of the set to be defined:
  - $CF = \{x \mid x \text{ is a cat friend of Chenli}\}$
  - $E = \{x \mid x \text{ is an even number greater than 3}\}$

*Exercise 1:*

Given the following sets:

|                            |                              |
|----------------------------|------------------------------|
| $A = \{a, b, c, 2, 3, 4\}$ | $E = \{a, b, \{c\}\}$        |
| $B = \{a, b\}$             | $F = \emptyset$              |
| $C = \{c, 2\}$             | $G = \{\{a, b\}, \{c, 2\}\}$ |
| $D = \{b, c\}$             |                              |

classify each of the following statements as true or false.

- |                   |                     |                         |
|-------------------|---------------------|-------------------------|
| (a) $c \in F$     | (d) $A \subseteq C$ | (g) $\{B\} \subseteq G$ |
| (b) $c \in E$     | (e) $D \subseteq E$ | (h) $D \subseteq G$     |
| (c) $\{c\} \in E$ | (f) $F \subseteq A$ | (i) $\{D\} \subseteq G$ |

*Exercise 2:*

Let  $A = \{a, b, c\}$ ,  $B = \{c, d\}$ ,  $C = \{d, e, f\}$ . Calculate the following:

- (a)  $A \cup B$
- (b)  $A \cap B$
- (c)  $A \cup (B \cap C)$
- (e)  $B \cup \emptyset$
- (g)  $A - B$

*Exercise 3 (from Mar 13<sup>th</sup>):*

Suppose our world contains the following sets:

$S = \{x: x \text{ swims}\}$   
 $D = \{x: x \text{ is a downhill skier}\}$   
 $B = \{x: x \text{ is a basketball player}\}$

The members of these sets are:

$U = \{\text{Ben, Christine, Dragan, Goran, Jonathan, Kai, Leo, Nancy, Susi, Uli}\}$   
 $S = \{\text{Ben, Christine, Jonathan, Kai, Leo, Nancy, Susi, Uli}\}$   
 $D = \{\text{Jonathan, Kai, Leo, Susi, Uli}\}$   $B = \{\text{Ben, Dragan, Goran, Uli}\}$

Determine whether each of the following statements is true or false.

- a.  $S \subseteq U$
- b.  $D \subseteq U$
- c.  $B \subseteq U$
- d.  $\{\text{Ben, Uli}\} \subseteq D$
- e.  $\{\text{Ben, Uli}\} \subseteq B$
- f.  $\{\text{Christine, Dragan}\} \subseteq D$
- g.  $\{\text{Christine, Dragan}\} \subseteq U$

## Semantics – Truth Value

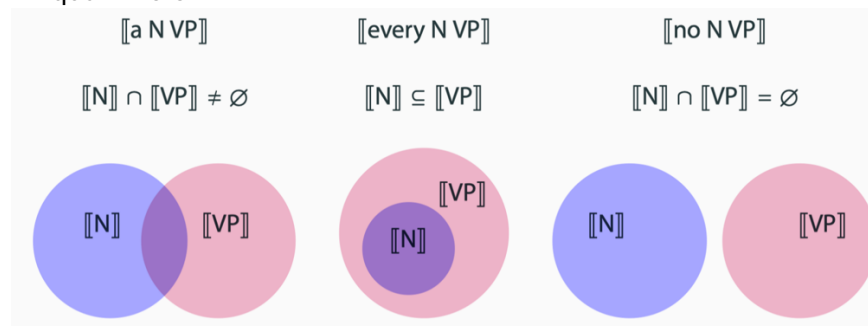
### 1. The extensions of ...

- **Sentence:** truth value (1 for true, 0 for false).
- **Name:** a single, specific individual/entity.
  - Example:  $\llbracket \text{Bart} \rrbracket = b$ ,  $\llbracket \text{Lisa} \rrbracket = l$ ,  $\llbracket \text{Maggie} \rrbracket = m$ .
- **Common noun:** a set of individuals.
  - Example:  $\llbracket \text{student} \rrbracket = \{b, l\}$
- **Adjective:** a set of individuals.
  - Example: Suppose Bart and Lisa are sleepy; then  $\llbracket \text{sleepy} \rrbracket = \{b, l\}$ .
- **Complex Noun Phrase:** the intersection of the sets denoted by the noun and adjective.
  - Example: Suppose Lisa and Maggie are happy and Bart and Lisa are students; then
  - $\llbracket \text{happy} \rrbracket = \{l, m\}$  and  $\llbracket \text{student} \rrbracket = \{b, l\}$ .
  - Then  $\llbracket \text{happy student} \rrbracket = \llbracket \text{happy} \rrbracket \cap \llbracket \text{student} \rrbracket = \{l, m\} \cap \{b, l\} = \{l\}$
- **Verb Phrase:** a set of individuals
  - Example: Suppose Lisa and Maggie cried; then  $\llbracket \text{cried} \rrbracket = \{l, m\}$ .
  - Example: Suppose Lisa and Bart go to school; then  $\llbracket \text{go to school} \rrbracket = \{l, b\}$ .

### 2. Semantic composition:

- **Complex Noun Phrase (adjective + noun):** the intersection of the sets denoted by the noun and adjective. Example given above.
- **Simple sentences (proper name + verb phrase):**
  - The meaning of a simple sentence in the form of 'NP VP':  $\llbracket \text{NP VP} \rrbracket = \llbracket \text{NP} \rrbracket \in \llbracket \text{VP} \rrbracket$

### 3. Sentences with quantifiers:



### Exercise 1:

Assume the following toy universe:

Alice, Bob, Clara, and Derek are **musicians**.

Alice, Clara, Edward, and Fiona are **painters**.

Bob, Derek, Fiona, and Grace are **college students**.

Grace, Clara, and Derek are **athletes**.

Provide the extensions for the following sets:

$\llbracket \text{musicians} \rrbracket$

$\llbracket \text{painters} \rrbracket$

Give the extension for the following using set theory:

Alice is a musician.

Edward is not a college student.

Clara is a painter and an athlete.

Some painters are musicians.

Every college student is an athlete.

No athlete is a painter.

Translate the following into English sentence:

$[[\text{athletes}]] \subseteq [[\text{college students}]]$

$[[\text{painters}]] \cap [[\text{musicians}]] \neq \emptyset$

## **Semantics and Pragmatics – Entailment vs Presupposition vs Implicature**

### 1. Entailment:

- For two sentences p and q, p entails q iff whenever p is true, q must also be true. There is no possible world such that p is true but q is false.
- “If p is true, does it automatically mean q? Is there a possible world that p is true while q is false?”
  - Example:
    - John ate an egg and a waffle for breakfast.
    - John ate a waffle for breakfast.
    - **Test:** If John ate an egg and a waffle for breakfast, it is automatically true that he ate a waffle. There is no possible world where he ate egg and waffle but did not eat waffle.
- **Vacuously true:** A sentence is vacuously true when it is true under any circumstances. A vacuously true sentence is entailed by any other sentences.
  - Example: John will have breakfast or he will not have breakfast.

### 2. Presupposition:

- A special type of entailment. For two sentences p and q, p presupposes q iff both p and the negation of p entail q.
- Test for presupposition:
  - **Negation test:** Negate the utterance. If the negated utterance still has the same inference, then that inference is a presupposition.
  - Example:
    - Utterance: John’s daughter left.
    - Inference: John has a daughter.
    - Negation test: the negated utterance ‘John’s daughter **didn’t** leave’ still has the inference that John has a daughter. Therefore, the inference ‘John has a daughter’ is a presupposition.

### 3. Implicature:

- The additional information that is inferred from speaker’s utterance.
- Test for Implicature:

- **Cancellation test: Deny** the inference and add it to the original utterance. If the resulting sentence is not contradictory, then the inference is an implicature.
  - Example:
  - Utterance: Dina speaks Odia, Hindi, and Telegu.
  - Inference: Dina speaks only these three languages.
  - Cancellation test: Dina speaks Odia, Hindi, and Telegu, but in fact she speaks other languages as well.
- **Reinforcement test: Affirm** the inference and add it to the original utterance. If the resulting sentence is not redundant, then the inference is an implicature.
  - Example:
  - Reinforcement test: Dina speaks Odia, Hindi, and Telegu, and she speaks only these three languages.
- **Scalar implicature:** a special type of implicature.
  - Stronger quantifier/larger number entail all weaker quantifiers/small numbers: all entails many, and half entails some; two entails one; three entails two and one
  - Weaker quantifiers/smaller number implicate the negation of stronger quantifiers/larger number: some implicates not all, not most, and not half; one implicates not two, not three, ...

4. A general strategy for testing whether an inference is an entailment, implicature, or presupposition:

- If the inference can be cancelled or reinforced, then that inference is an implicature.
- If the inference persists when the sentence is negated, then the inference is a presupposition.
- Otherwise, the inference is an entailment.

*Exercise 1:*

Determine the type of inferences in (a) and (b): is it an entailment, implicature, or presupposition?

1. Mary is playing soccer again.

- a. Mary plays soccer now.
- b. Mary has played soccer before.

2. The fact that John graduated delighted Mary.

- a. John graduated.
- b. The fact that John graduated caused joy in Mary.

3. Jack has two children.

- a. Jack does not have more than two children.
- b. Jack has one child.

4. Amy used to swim.

- a. Amy has swam before.
- b. Amy no longer swims.

5. John's daughter got married and moved to NY.
- a. John has a daughter.
  - b. John's daughter got married first and then moved to NY.
6. Mary also owns a car.
- a. Mary owns exactly one car.
  - b. Someone else owns a car, too.

### **Pragmatics - Grice's Maxim**

#### 1. Maxim of Quantity:

- Make your contribution as informative as it is required.
- Do not make your contribution more or less informative than is required.
- Example:
  - Q: What language does Dina speak?
  - A: She speaks Odia, Hindi and Telugu.
  - **Flouting:** She actually speaks French as a fourth language.
  - **Implicature:** She only speaks these three languages.

#### 2. Maxim of Quality:

- Do not say what you believe to be false.
- Do not say that for which you lack adequate evidence.
- Example:
  - Q: Was it raining on Tuesday in New Jersey?
  - A: No, it was sunny
  - **Flouting:** it was actually raining on Tuesday.

#### 3. Maxim of Relevance:

- Be relevant.
- Example:
  - Q: What do you want for dinner?
  - **Flouting:** I made the answers for the quiz last night.

#### 4. Maxim of Manner:

- Avoid obscurity of expression and ambiguity.
- Be brief.
- Be orderly.
- Example:
  - RJ got a green scarf as a gift and stopped complaining.
  - **Flouting:** RJ's complaints weren't relevant to the gifted green scarf.
  - **Implicature:** RJ was complaining that he didn't have a green scarf.

### **Phonetics**

**Important to remember:**

1. There are three dimensions used to characterize consonants

- **Place of articulation:** where the obstruction of airflow occurs
- **Manner of articulation:** how the obstruction of airflow occurs
- **Voicing:** whether the vocal fold vibrates

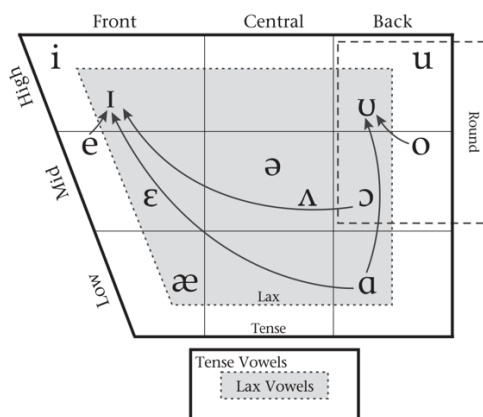
2. There are four dimensions used to characterize vowels

- **Tongue height:** high, mid or low
- **Tongue advancement:** [+front,-back], [-front,-back], [-front,+back]
  - In other words, front, central or back
- **Lip rounding:** rounded or unrounded
- **Tenseness:** tense or untensed/lax

3. There are complex consonants (affricates) and complex vowels (diphthongs), which are essentially two consonants or vowels put together, involving transition from one sound to the other.

4. **Natural class:** A group of sounds is a natural class if there's an articulatory description that picks out those and only those sounds – no more, no less.

(4) Two-part articulations of the diphthongs of English (the arrows indicate the transitions)



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### Exercise 1

Give the IPA symbol that corresponds to each articulatory description:

1. Voiceless bilabial plosive: \_\_\_\_\_
2. Voiced alveolar fricative: \_\_\_\_\_
3. Voiced velar nasal: \_\_\_\_\_
4. Voiceless glottal plosive: \_\_\_\_\_

### Exercise 2

State whether each of the following groups of sounds constitute a natural class (in English). If they do, state the articulatory descriptions that describe that natural class.

1. {f θ ʃ s h}
2. {d n r z ɹ l}

3. {b d g j}

### Exercise 3

Write the four-part articulatory descriptions for the sounds represented by the symbols.

- a. [ʌ] \_\_\_\_\_ b. [ɪ] \_\_\_\_\_  
c. [u] \_\_\_\_\_ d. [ɔ] \_\_\_\_\_

### Exercise 4

Match the following English words (given in alphabetical order) to the transcriptions

*added, aided, books, box, bucks, cloth, clothe, reach, rich, ridge, these, this*

- a. [rɪtʃ] \_\_\_\_\_ b. [rɪtʃ] \_\_\_\_\_  
c. [rɪdʒ] \_\_\_\_\_ d. [bʌks] \_\_\_\_\_  
e. [bʊks] \_\_\_\_\_ f. [bʌks] \_\_\_\_\_  
g. [eɪdəd] \_\_\_\_\_ h. [ædəd] \_\_\_\_\_  
i. [ðɪz] \_\_\_\_\_ j. [ðɪs] \_\_\_\_\_  
k. [klaʊð] \_\_\_\_\_ l. [klɒθ] \_\_\_\_\_

### Exercise 5

Correct the mistakes in the following phonetic transcriptions of English words

- a. [ʃut] shut  
b. [lɜft] left  
c. [tʃɪld] child  
d. [swɪt] swift

## Phonology – Phonemes and Allophones

1. A **minimal pair** is a pair of words that differ only in one sound and have different meanings.

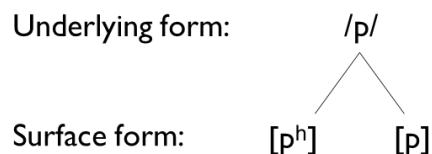
- If two sounds switch places, that counts as two differences.
- Go through the data thoroughly. But if you can't find a minimal pair after a thorough search, then believe in yourself and just say there's no minimal pair.

2. If there exist a minimal pair for the two sounds, then the two sounds are **contrastive**.

- The contrasts are defined with respect to a specific language.

3. The non-contrastive sounds in a language are often treated as special realizations (**allophones**) of the same phoneme.

- The allophones are the surface realizations of the underlying phoneme. Consider aspiration in English:



- Allophones (phonetic representations) are expressed in [ ]; while the phonemes (phonemic representations) are expressed in / /.



- Allophones appear in **complementary distributions**, meaning that they never appear in the same environment.

Exercise (from reading):

1. *Inuktitut* (Eastern) (Native Canadian)

The data here are in a relatively broad transcription.

|             |                         |           |                       |
|-------------|-------------------------|-----------|-----------------------|
| a) iglumut  | 'to a house'            | h) pinna  | 'that one up there'   |
| b) ukiaq    | 'late fall'             | i) ani    | 'female's brother'    |
| c) aiviq    | 'walrus'                | j) iglu   | '(snow)house'         |
| d) aniguvit | 'if you leave'          | k) panna  | 'that place up there' |
| e) aglu     | 'seal's breathing hole' | l) aivuuq | 'she goes home'       |
| f) iglunit  | 'from a house'          | m) ini    | 'place, spot'         |
| g) anigavit | 'because you leave'     | n) ukiuq  | 'winter'              |

- i) List all the minimal pairs in this data. Based on the minimal pairs you have found, list all the contrastive pairs of vowels. (Refer to Section 1.1.)

- iii) Now consider the data again; here it is transcribed in narrower transcription. In it, there are phonetically similar segments that are in complementary distribution. Look for them and then answer the question that follows the data.

|              |                         |            |                       |
|--------------|-------------------------|------------|-----------------------|
| aa) iglumut  | 'to a house'            | hh) pinna  | 'that one up there'   |
| bb) ukiaq    | 'late fall'             | ii) ani    | 'female's brother'    |
| cc) aiviq    | 'walrus'                | jj) iglu   | '(snow)house'         |
| dd) aniguvit | 'if you leave'          | kk) panna  | 'that place up there' |
| ee) aglu     | 'seal's breathing hole' | ll) aivuuq | 'she goes home'       |
| ff) iglunit  | 'from a house'          | mm) ini    | 'place, spot'         |
| gg) anigavit | 'because you leave'     | nn) ukiuq  | 'winter'              |

- iv) List the phonetically similar segments that are in complementary distribution. State their distribution in words. (Refer to Sections 2 and 3.)

# 10. English

Many speakers of English have two variants of [l]. One, called *clear l*, is transcribed as [l] in the following data. The other, called *dark l*, is transcribed with [ɫ]. Use of [l] and [ɫ] varies according to dialect. Examine the data from one dialect, and answer the questions that follow.

|             |           |                        |        |
|-------------|-----------|------------------------|--------|
| a) [laɪf]   | 'life'    | g) [p <sup>h</sup> ɪɫ] | 'pill' |
| b) [lɪp]    | 'leap'    | h) [fiɫ]               | 'feel' |
| c) [luːz]   | 'lose'    | i) [heɪp]              | 'help' |
| d) [ɪlowp]  | 'elope'   | j) [bʌɫk]              | 'bulk' |
| e) [dɪlaɪt] | 'delight' | k) [sowɫd]             | 'sold' |
| f) [slɪp]   | 'sleep'   | l) [fʊɫ]               | 'full' |

Do [l] and [ɫ] belong to separate phonemes or are they allophones of the same phoneme? If you think they belong to separate phonemes, answer question *i*. If you think they are allophones of the same phoneme, answer questions *ii-iv*. (Refer to Section 2.)

- List the evidence that makes your case for considering [l] and [ɫ] as separate phonemes.
- State the distribution of [l] and [ɫ] in words.
- Which variant makes the best underlying form? Why?